Experiment No: 6

OBJECTIVE: - Verification of state tables of RS, JK, T and D Flip-Flop using NAND & NOR gates.

APPARATUS REQUIRED:-

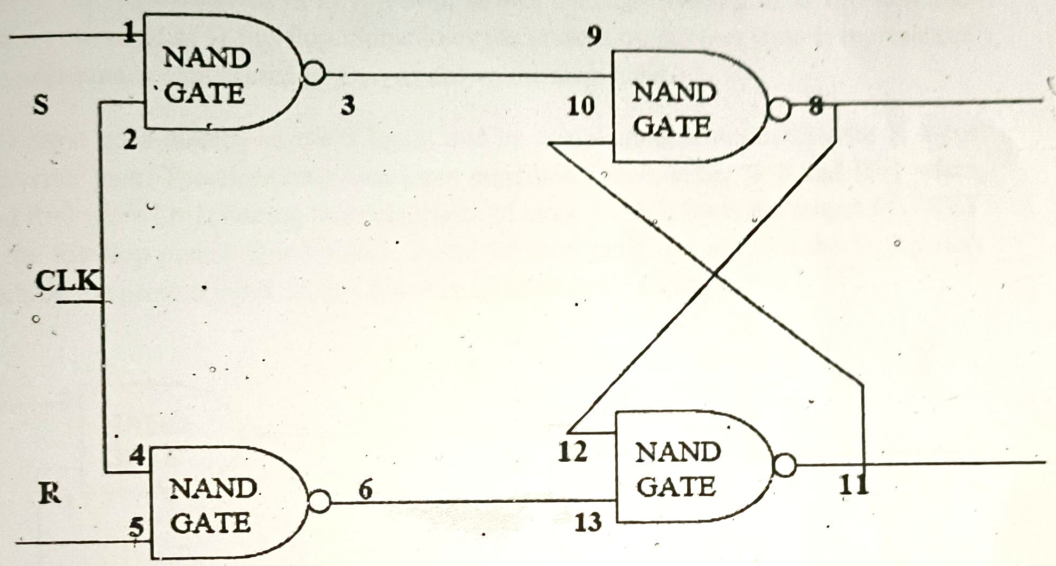
S.No.	APPARATUS	SPECIFICATION	QUANTITY
1.	NAND Gate	IC 7400	02
2.	NOT Gate	IC 7404	01
3.	IC Trainer Kit	-	01
4.	Patch Cords	-	As Required

THEORY: - Flip-flop is a bistable multivibrator. A flip-flop has only two stable states. A flip-flop is abbreviated as FF. The basic unit of memory is the latch. The output of the flip-flop is either logic 0 or logic 1. This means that it can be store 1-bit information, so it is called 1-bit memory unit.

The basic RS latch is an asynchronous transparent sequential circuit. This means that any change in the output of R or S is transmitted immediately to the output at Q and $\bar{Q}$ according to the truth table. The operation of the basic flip-flop (latch) can be modified by providing an additional control input that determines when the state of the circuit is to be changed. This additional control input is called 'clock' or 'clock pulse'. With this clock, the flip flop is called clocked flip-flop or synchronous flip-flop. The term synchronous indicates that the output changes its state only at a specified point or time according to an input the changes in the output occur in synchronous with the clock.

S-R FLIP-FLOP:- R-S Flip Flop is one that has two inputs R & S and two outputs Q & $\bar{Q}$. A R-S Flip Flop can be constructed by using NOR gates or NAND gates. R-S Flip Flop is constructed by using four NAND gates. It is not always desirable to allow a latch to change states at random times. In synchronous operation one pair of gates are connected as an RS flip-flop. A second pair; called steering gates, can be enabled by clock signal. This clock signal allows one or the other of these gates to pass a SET or RESET signal to the flip-flop.

The S and R inputs are free to change at random, but the flip-flop output will change only when the clock is high. If the clock is LOW, the flip-flop is disabled even if all the inputs appear at the input terminal. Shows the clocked RS flip-flop. Table shows the truth table of clocked RS flip-flop.



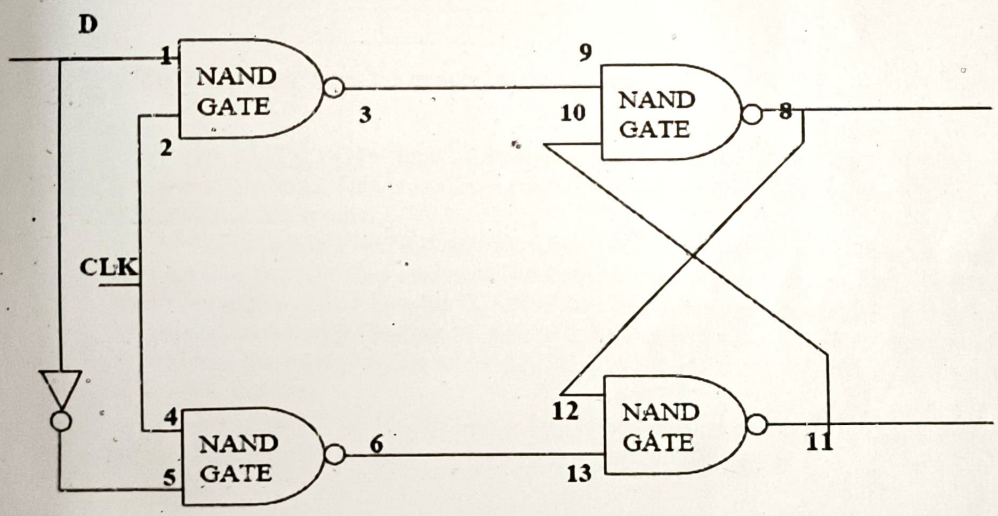
Logic Diagram for S-R Flip-Flop using NAND Gate

Input			Outputs
CLK	S	R	Q(t+1)
0	X	X	No Change
1	0	0	No Change
1	0	1	0
1	1	0	1
1	1	1	FORBIDDEN

Truth Table of S-R Flip-Flop

FLIP-FLOP:- The output of RS flip-flop is an unpredictable state when both the inputs are some. (Inputs equal to 00 means no change and 11 means unpredictable state) These two conditions are not required for particular application. These input conditions can be avoided by making them complementary to each other. The modified clocked RS flip-flop is known as D Flip-flop. Shows the logic diagram of D flip-flop. Shows the logic symbol of D flip-flop and table shows the truth table of D flip-flop. Sometimes the present output (or) state is represented by Q_n and the next state is represented by Q_{n+1} as shown the truth table.

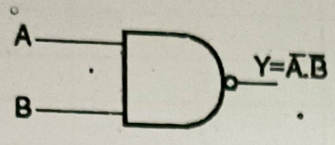
The D input goes directly to the S input, and its complement is applied to the R input through an inverter gate. Therefore only two input conditions exist, either S=0 and R=1 when D=0, S=1 and R=0 when D=1. During this occurrence of clock pulse if D=1, the output Q is SET and if D=0, the flip-flop output Q is RESET. Form the truth table, we see that the D flip-flop output depends on the present input, so this flip-flop is called delay flip-flop.



Logic Diagram for D Flip-Flop using NAND Gate

Input		Outputs	
CLK	D	Q_n	Q(n+1)
1	0	1	0
1	1	0	1

Truth Table of D Flip-Flop

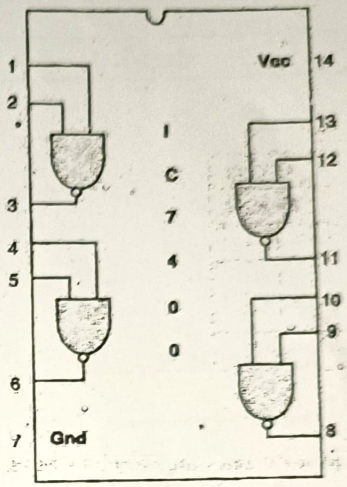


SYMBOL

Inputs		Output
A	B	$Y=\overline{AB}$
0	0	1
0	1	1
1	0	1
1	1	0

TRUTH TABLE FOR A 2 INPUTS NAND GATE

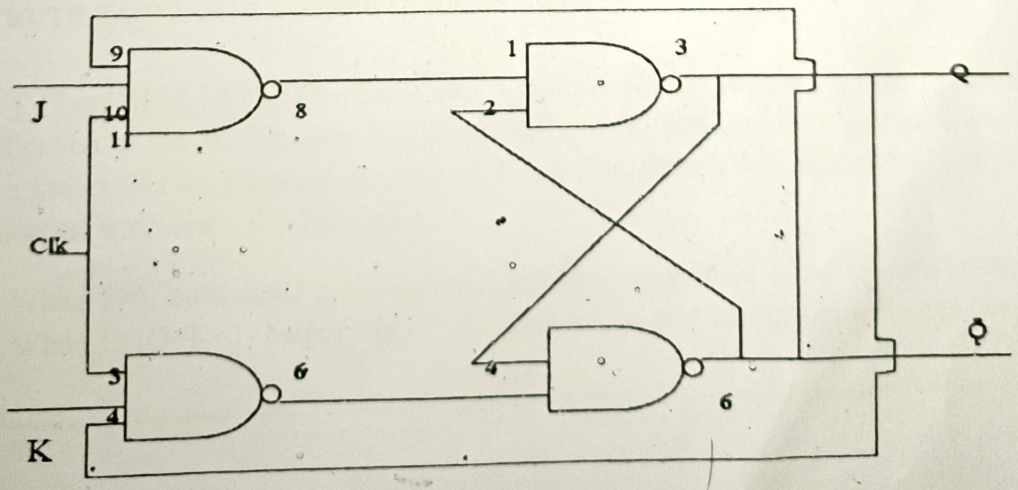
PIN DIAGRAM



JK FLIP-FLOP:- Inputs Setting R=S=1 with an R-S Flip Flop, Q (t +1) and Q (t +1) to the same logic level. This is an illegal conditions .The J-K Flip Flop account for this illegal input. It is used to build counter.

The values of J and K determine what a J-K Flip Flop dose on the next clock edge. When both are low, the Flip Flop retains its last state. When J is low and K is high, the Flip Flop resets. When J is high and K is low, the Flip Flop sets, When both are high the Flip flop toggles. In this last mode, the J-K Flip Flop can be used as a frequency divider. It is a refinement of the RS flip-flop. Shows the logic diagram of JK flip-flop. Shows the logic symbol of JK flip-flop and truth table of JK flip-flop.

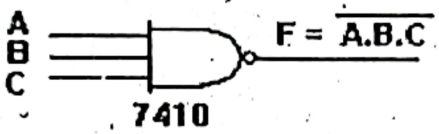
The Q output of JK flip-flop is connected back to one of the inputs of gate 1. The $\overline{Q}$ output of JK flip-flop is connected back to the input of the gate 2.



Logic Diagram for JK Flip-Flop using NAND Gate

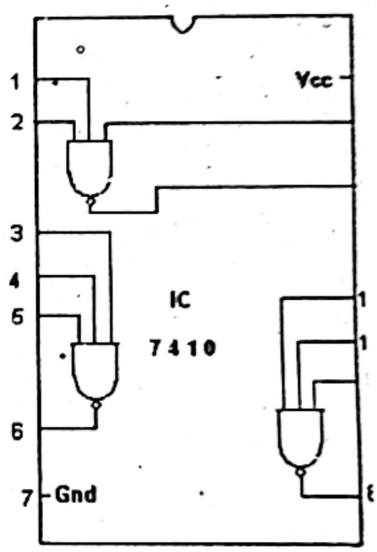
Input			Outputs
CLK	J	K	Q(t+1)
0	X	X	No Change
1	0	0	No Change
1	0	1	0
1	1	0	1
1	1	1	Toggle

Truth Table of JK Flip-Flop



SYMBOL

PIN DIGRAM



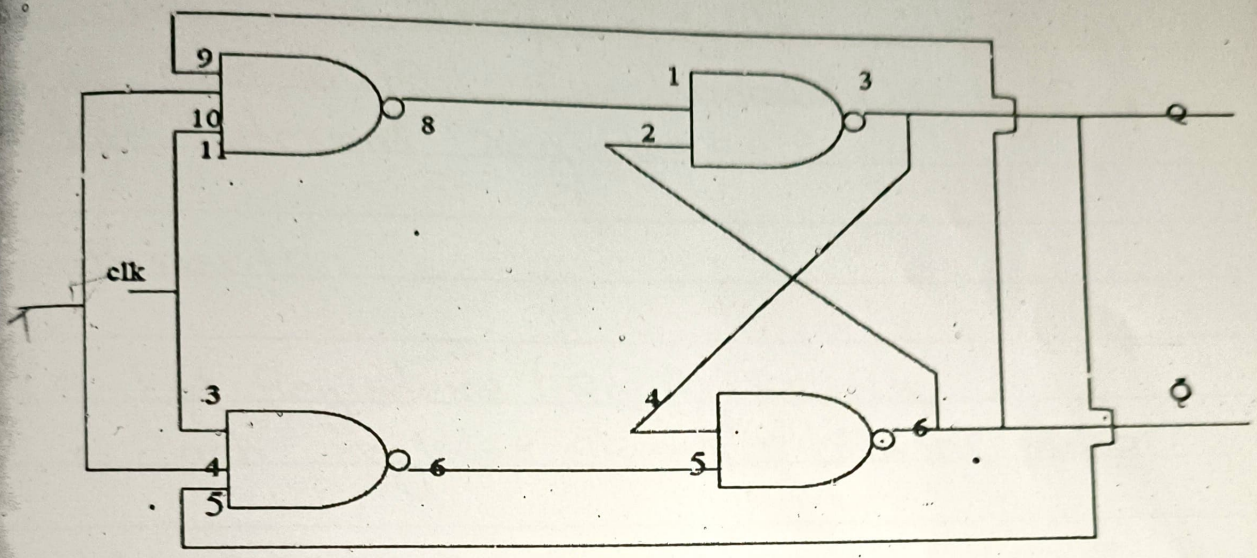
Inputs			Output
A	B	C	Y= A.B.C
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

TRUTH TABLE FOR A 3 INPUTS NAND GATE

T (Toggle) FLIP-FLOP:- The T flip-flop is known as 'Toggle flip-flop'. The flip-flop is a modification of the JK flip-flop. It is obtained from JK flip-flop by connecting both inputs J and K together. Shows the logic diagram of T flip-flop. Shows the logic symbol of T flip-flop and shows the truth table of T flip-flop.

. When T=0, both AND gates are disabled and hence there is no change in the previous output. When T=1(J=K=1) output toggles. Toggles means that the output is 0 when the previous

state is 1 otherwise output is 1 when the previous state is 0. So the output is a complement of the previous output.



Logic Diagram for T Flip-Flop using NAND Gate

PROCEDURE:-

1.) Connections are given as per circuit diagram.
2.) Logical inputs are given as per circuit diagram.
3.) Observe the output and verify the truth table.

RESULT: - Truth table has been verified.

PRECAUTIONS:-

1.) All Connections should be according to circuit diagram.
2.) All Connections should be right and tight.
3.) Reading should be taken carefully.
4.) Powers supply Switched Off after completing the Experiment

Experiment- 7

Expt. No. ---------- Date ----------

EXPERIMENT No. 7

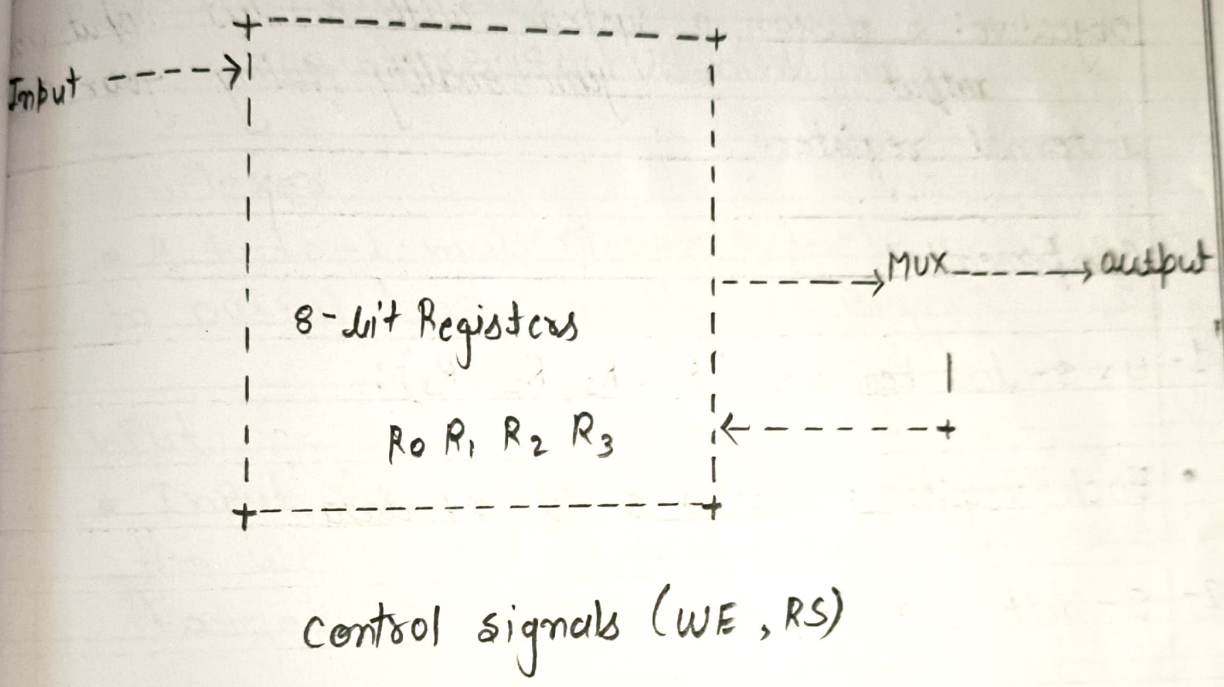
Page No. 01

objective: To design a system with 8-bit input and output functionality using four 8-bit internal registers.

Components:

1. 4 x 8-bit Registers (R0, R1, R2, R3)
   - Each register stores 8 bits of data.
2. 8-bit Data bus
   - Used to carry input data to the system and output data from the system.
3. Multiplexer (MUX)
   - Selects data from one of the four registers based on a 2-bit select signal
4. Control Signals
   - write Enable (WE) Enables writing data to selected register
   - Register Select (RS) 2-bit signal to choose which register to read/write (R0, R1, R2, or R3)

Teacher's Signature : _____


Input
MUX
output
8-bit Registers
R0 R1 R2 R3
control signals (WE, RS)

Expt. No. _____ Date _____

Page No. 03

Design overview

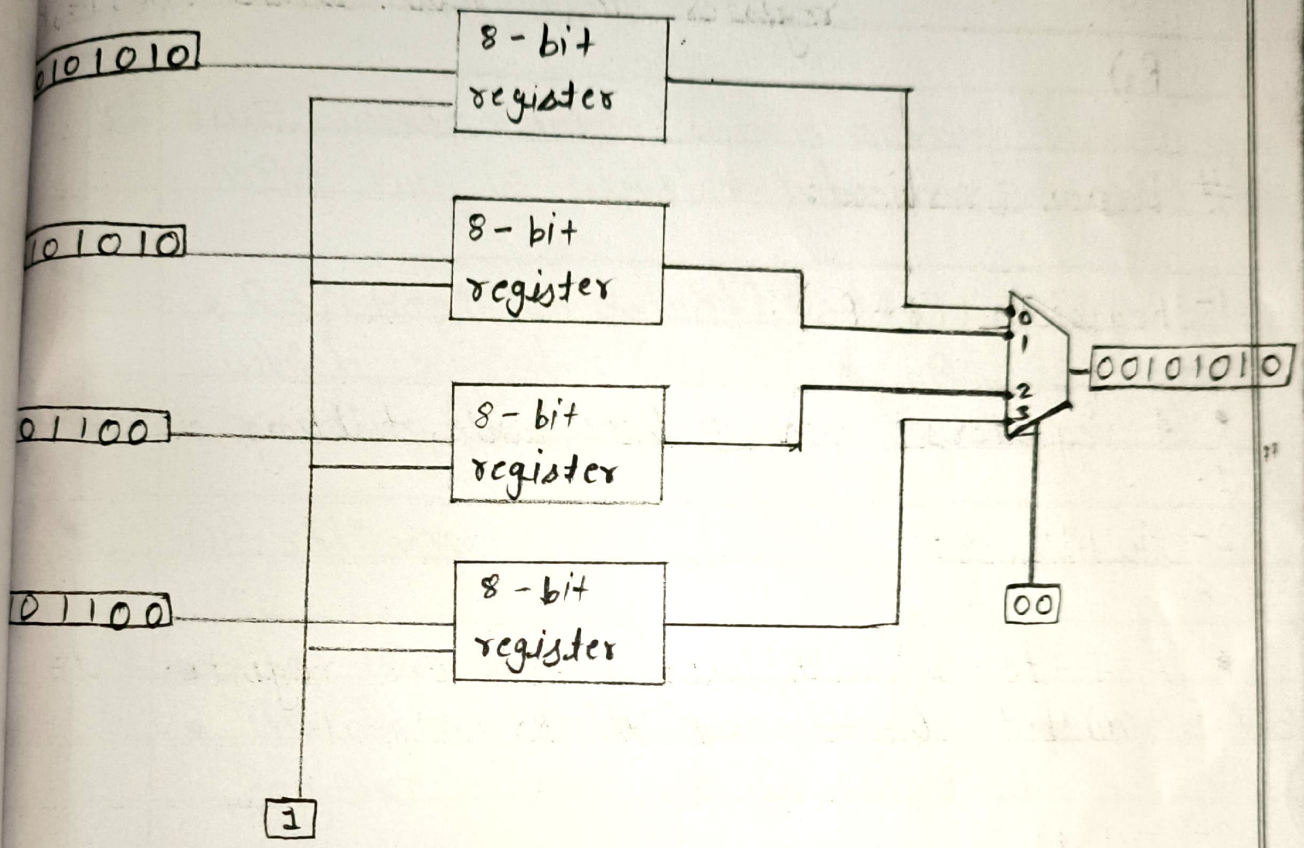
1. **Registers (R0 – R3)**
   - 4 registers each 8 bits wide holding data

2. **Multiplexer**
   - A 4-to-1 multiplexer selects one register to output based on the RS signal.

3. **Input**
   - Input data is written to a register when the WE signal is active and RS selects the target register.

4. **Output**
   - The output data is selected from the Registers via the MUX controlled by the RS signal.

Teacher's Signature : _____


8 - bit register
8 - bit register
8 - bit register
8 - bit register
0 1 0 1 0 1 0
1 0 1 0 1 0
0 1 1 0 0
1 0 1 1 0 0
0 1 2 3
0 0 1 0 1 0 1 0
0 0
1

Expt. No. _____ Date _____

Page No. 34

Control Signals

- WE (write Enable) when active, data from the input bus is written into the selected registe
- RS (Register Select) 2-bit signal to choose which register (R_0, R_1, R_2, R_3) is involved in reading/writing.

Operations

1 write operation

- The input data is written to the selected Register when WE is Enabled.

Read Operation

- The MUX selects one of the registers base on the RS signal, and the output bus carries the data from the selected registe

Summary

The system uses four 8-bit registers for storing data, with a multiplexer to select which register's data is sent to the outpu Control signals manage writing to and

Teacher's Signature : 4/11/24

Experiment-8

EXPERIMENT: Implementation of Arithmetic Logic Unit.

APPARATUS REQUIRED:

S.No.	Equipment	Qty.
1	General purpose digital trainer	1
2	IC-74181	1

THEORY: - Arithmetic logic unit is a multipurpose device capable of providing several different arithmetic and logic operations. The specific o peration to be performed is selected by the user by placing a specific binary code on the mode select i/p. ALU s are available in large scale integrated circuit packages.

Functional block diag. For 74181 ALU is shown in fig. It is a 4-bit ALU, which provides 16 arithmetic plus 16 logic operations. The unit accepts two 4-bit words (A_3 A2 A_1 A0 and B_3 B_2 B_1 B_0) and a carry i/p Cn as i/p's.

The operation to be performed on these i/p are determined by logic levels on i/ps

PROCEDURE:

1. Put IC on the breadboard.
2. Apply Vcc supply at pin 24.
3. Apply ground at pin 12.
4. Make connections as shown in the circuit diagram.
5. Observe the different outputs.

OBSERVATION TABLE:

SELECTION				M=1 LOGIC FUNCTION	M=0 ARITHMRTIC OPERATION
S_3	S_2	S_1	S_0		
0	0	0	0	F=A	F=A
0	0	0	1	F=A+B	F=A+B
0	0	1	0	F=AB	F=A+B
0	0	1	1	F=0	F= -1
0	1	0	0	F=AB	F=A+AB
0	1	0	1	F=B	F=(A+B)+ AB
0	1	1	0	F=A O B	F=A – B -1
0	1	1	1	F=AB	F=AB -1

1	0	0	0	F=A+B	F=A + AB
1	0	0	1	F=A O B	F=A+B
1	0	1	0	F=B	F= (A+B) + AB
1	0	1	1	F=AB	F=AB -1
1	1	0	0	F=1	A + A*
1	1	0	1	F=A+B	F=(A+B) +A
1	1	1	0	F=A+B	F=(A+B) +A
1	1	1	1	F=A	F=A - 1

RESULT: The functional table is verified for 74181 IC.

PRECAUTIONS: -

1. Connection should be tight.
2. O/P should be finding sequentially.
3. IC's should be handled carefully.

Truth Table (SISO)

Input		**Output**
tn		**tn+8**
Ds1	Ds2	Q7
L	L	L
L	H	L
H	L	L
H	H	H

Truth Table (SIPO)

Operating mode	**Input**			**Output**	
	MR	**DS1**	**DS2**	**Q0**	**Q1 - Q7**
Reset	L	X	X	L	L - L
Shift	H	L	L	L	q0 - q6
	H	H	H	L	q0 - q6
	H	H	H	L	q0 - q6
	H	H	H	H	q0 - q6

Truth Table (PIPO)

Operating mode	Input MR	S1	S0	DSR	DSL	PN	Output Q0	Q1	Q2	Q3
Reset	L	X	X	X	X	X	L	L	L	L
Hold	H	L	L	X	X	X	q0	q1	q2	q3
Shift left	H	H	L	X	L	X	q1	q2	q3	L
	H	L	H	X	L	X	q1	q2	q3	H
Shift right	H	L	H	L	X	X	L	q0	q1	q2
	H	L	H	H	X	X	H	q0	q1	q2
Parallel load	H	H	H	X	X	Pn	P0	P1	P2	P3

RESULT:- All Shift Register Verify With the given truth table.

PRECAUTIONS: -

1. Connection should be tight.
2. O/P should be finding sequentially.
3. IC's should be handled carefully.